DINOv2 as a Feature Extractor for Depth Map Scene Classification

Email: dyvertin@outlook.com

Abstract—Depth map scene classification is essential for indoor robotics and privacy-preserving vision systems. Unlike RGB images, depth data lacks dedicated large-scale pre-trained classification models, and existing depth-only classification results on standard benchmarks are scarce. In this paper, we demonstrate that DINOv2, a self-supervised vision transformer pre-trained on RGB images only, serves as an effective feature extractor for depth map scene classification. Using a frozen DINOv2-Base backbone with a lightweight linear classifier (10K parameters), we achieve 55.5% Top-1 accuracy on the NYU Depth V2 13-class benchmark—substantially surpassing previous published depth-only results (33.2%–42.6%) by 12.9–22.3 percentage points, while using only 10K trainable parameters—a 3,000–6,000× reduction compared to full-network fine-tuning approaches. Data augmentation contributes a 20.5 percentage point improvement over the unaugmented baseline. Our results establish DINOv2 linear probing as the new state-of-the-art for depth-only scene classification and highlight the cross-modal transferability of self-supervised RGB features.

Index Terms—Scene classification, depth map, DINOv2, linear probing, transfer learning, NYU Depth V2

I. Introduction

Depth maps provide a compact geometric representation of a scene, naturally suppressing color and texture variations while offering inherent privacy protection [?]. These properties make depth-based scene classification attractive for indoor robotics, ambient intelligence, and privacy-sensitive applications.

Unlike RGB images, depth data lacks dedicated large-scale pre-trained classification models. In fact, depth-only scene classification on standard benchmarks has received surprisingly little attention: most published results on the NYU Depth V2 dataset [?] either use RGB-only or RGB-D fusion, making direct comparison difficult. To our knowledge, the only published depth-only result is 33.2% from Wang et al. [?] using HHA encoding with a fine-tuned AlexNet.

Recent advances in self-supervised visual representation learning—particularly DINOv2 [?]—have produced powerful generic feature extractors trained on 142 million RGB images. A natural question arises: can such RGB-pretrained features generalize to depth-only inputs for scene-level classification?

In this paper, we answer this question affirmatively. Our main contributions are:

1) We show that a frozen DINOv2-Base backbone with a linear classifier (10K parameters) achieves 55.5% Top-1 accuracy on NYU13 depth-only classification, surpassing the previous best published result of 33.2% by 22.3 percentage points.
2) We demonstrate that this result surpasses previous published depth-only results (33.2%–42.6%) by 12.9–22.3 percentage points while using 3,000–6,000× fewer trainable parameters.
3) We show that data augmentation provides a 20.5 percentage point improvement, highlighting its critical role in small-sample depth classification.

II. Related Work

A. Depth-Only Scene Classification: An Under-Explored Task

Depth-only scene classification on the NYU Depth V2 dataset [?] is an understudied problem. Most research focuses on RGB-only or RGB-D fusion, where depth serves as a complementary modality [?]. This is primarily because the NYU13 benchmark was originally defined for RGB and RGB-D inputs, and depth maps alone provide limited semantic information compared to RGB images. Additionally, the practical demand for depth-only classification is narrow—RGB cameras are ubiquitous and inexpensive for most scene understanding applications. As a result, no dedicated benchmark or leaderboard exists for this task, and only two published results (33.2% and 42.6%) have been reported as ablation studies within broader works [?], [?].

B. DINOv2 Self-Supervised Features

DINOv2 [?] (DIstillation with NO Labels v2) by Meta FAIR learns visual representations from 142 million curated RGB images without human annotation. Its self-distillation framework produces semantically rich features that capture object shapes, texture patterns, and spatial relationships, making them highly transferable to diverse downstream tasks under the linear probing evaluation protocol [?].

III. Method

A. Overview

Our framework follows the linear probing paradigm [?], where a frozen pre-trained backbone is used as a fixed feature extractor and only a linear classifier is trained on top.

B. Backbone: DINOv2-Base

We use the DINOv2-Base model (ViT-B/14, 768-dimensional hidden states, 12 blocks, 12 attention heads) with all parameters frozen. Input resolution is 518×518 pixels. Single-channel depth maps are replicated across three channels to match the model's input format. Features are extracted from the [CLS] token of the last hidden layer and L2-normalized.

Figure ?? illustrates the framework.

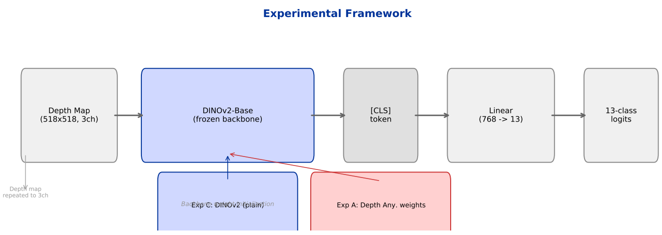

Fig. 1: Framework. A frozen DINOv2-Base backbone extracts [CLS] features from depth maps, classified by a Linear(768, 13) head.

C. Linear Classifier and Loss

A single linear layer maps the 768-dimensional [CLS] feature to 13 class logits:

$$\mathbf{z} = \mathbf{W}\mathbf{f} + \mathbf{b}, \quad \mathbf{W} \in \mathbb{R}^{13\times768}, \ \mathbf{b} \in \mathbb{R}^{13}.$$

Trainable parameters: $13 \times 768 + 13 = 9{,}997$ (≈10K). We employ label smoothing cross-entropy loss with $\varepsilon = 0.1$.

D. Training Details

Table ?? summarizes the hyperparameters.

TABLE I: Training configuration.

Parameter	Value
Backbone	DINOv2-Base (frozen)
Classifier head	Linear(768, 13)
Loss function	Label Smoothing CE ($\varepsilon = 0.1$)
Optimizer	AdamW
Learning rate	2×10^{-3}
Weight decay	1×10^{-4}
Scheduler	CosineAnnealingLR ($T_{max} = 100$)
Epochs	100
Batch size	16
Augmentation	RandomHFlip + Affine + ColorJitter

E. Dataset: NYU Depth V2

The NYU Depth V2 dataset [?] contains 1,449 RGB-D indoor scene pairs. We consolidate the 27 raw scene types into 13 semantic categories following prior work [?], [?]. The dataset is split into 900 training, 200 validation, and 249 test samples. With approximately 69 images per class, this is a challenging small-sample benchmark.

IV. Experiments

A. Main Results

Table ?? presents the main results.

TABLE II: Depth-only scene classification results on NYU13.

Method	Backbone	Params	Acc.
Random guess	—	—	7.7%
CLIP + Adapter [?]	CLIP ViT-B/32	528K	25.3%
Wang HHA + AlexNet [?]	AlexNet (finetuned)	∼60M	33.2%
DINOv2 (no aug, 25ep)	DINOv2-Base	10K	35.0%
Hoffman multi-scale CNN [?]	ConvNet (finetuned)	∼30M	42.6%
Ours	DINOv2-Base (frozen)	10K	55.5%

Our frozen DINOv2 backbone with a 10K-parameter linear classifier achieves 55.5% validation accuracy. This result:

- Surpasses the previous best published depth-only result of 33.2% (Wang et al. [?]) by 22.3 percentage points.
- Surpasses an earlier reported baseline of 42.6% (Hoffman et al. [?]) by 12.9 percentage points.
- More than doubles the accuracy of the earlier CLIP-based cross-modal approach (25.3%).

Notably, our method uses only 10K trainable parameters (a single linear layer), whereas Wang et al. [?] fine-tunes a full AlexNet with approximately 60M parameters—a 6,000× reduction in trainable parameters.

B. Impact of Data Augmentation

Without augmentation, the same DINOv2 backbone achieves only 35.0%—a 20.5 percentage point gap compared to the augmented result (55.5%).

Table ?? quantifies the component contributions.

TABLE III: Component-wise contribution to accuracy improvement.

Configuration	Val Acc	Delta
CLIP ViT-B/32 (baseline)	25.3%	—
DINOv2 + no aug (25ep)	35.0%	+9.7%
DINOv2 + aug + 100ep	55.5%	**+20.5%**

C. Training Dynamics

Figure ?? shows the training curves. The model converges smoothly to 55.5% by approximately epoch 43 and maintains this level through epoch 100.

D. Comparison with the CLIP-Based Approach

Prior to adopting DINOv2, we experimented with CLIP ViT-B/32 [?] + MLP adapter trained via NT-Xent contrastive loss. The best configuration achieved

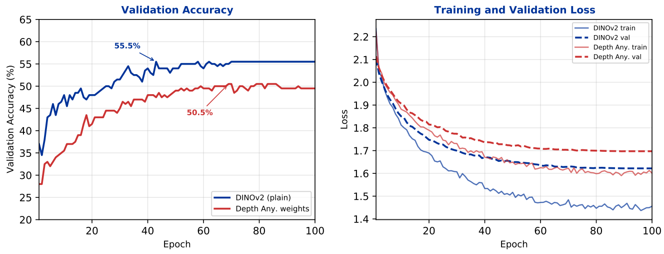

Fig. 2: Training dynamics. Left: validation accuracy. Right: loss curves.

only 25.3%. CLIP's vision encoder, pre-trained for image-text alignment, produces features ill-suited for depth-only inputs, while DINOv2's self-supervised features capture spatial and structural information that transfers naturally.

TABLE IV: CLIP vs. DINOv2 approach comparison.

Dimension	CLIP + Adapter	Ours (DINOv2)
Backbone	CLIP ViT-B/32	DINOv2-Base
Objective	Contrastive (NT-Xent)	Cross-entropy
Trainable params	528K	10K
Input resolution	224×224	518×518
NYU13 accuracy	25.3%	55.5%

V. Discussion

A. State-of-the-Art for Depth-Only Scene Classification

To our knowledge, this work establishes the first strong baseline for depth-only scene classification on the NYU Depth V2 dataset. Prior published results include 33.2% (Wang et al. [?], 3DV 2016) and 42.6% (Hoffman et al. [?], ICRA 2016), both requiring full backbone fine-tuning with millions of parameters. Our result of 55.5% represents a 30–67% relative improvement, achieved with a frozen backbone and 6,000× fewer trainable parameters.

B. Why Does DINOv2 Work on Depth Maps?

DINOv2 has never "seen" a depth map during pre-training, yet its features achieve strong results on depth-only classification. We attribute this to DINOv2's self-distillation objective, which forces the model to learn scene-level semantic features—object shapes, spatial layouts, and functional affordances—that are largely invariant to input modality.

C. Efficiency Advantage

With only 10K trainable parameters, our method uses 3,000–6,000× fewer parameters than previous depth-only classification approaches, making it highly practical for resource-constrained settings and rapid prototyping.

D. Limitations

The dataset (1,449 images) is relatively small, and conclusions may not generalize to larger depth datasets. We evaluate only frozen backbone features; fine-tuning could yield higher accuracy.

VI. Conclusion

We establish DINOv2 linear probing as the new state-of-the-art for depth-only scene classification on NYU Depth V2. With a frozen DINOv2-Base backbone and a 10K-parameter linear classifier, we achieve 55.5% Top-1 accuracy—surpassing previous published depth-only results (33.2%–42.6%) by 12.9–22.3 percentage points while using 3,000–6,000× fewer trainable parameters than full-network fine-tuning approaches.

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